Your TA will lend your group a wooden model of a solid $S \subseteq \mathbb{R}^3$ and share the equations that define the solid S. Use these to solve all problems on this worksheet.
Please do NOT draw on the model and return it to your TA at the end of tutorial.

T15.1 Use your model to understand the relationship between the geometry and the algebra of the solid S.

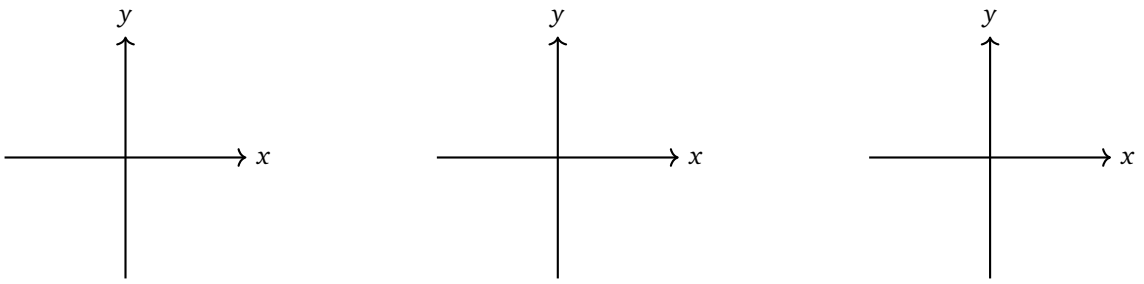
(a) Which parts of ∂S does each equation correspond to?

- The equation _____ corresponds to the flat bottom.
- The equation _____ corresponds to the slanted top.
- The equation _____ corresponds to the rounded side.

(b) For $z \in \mathbb{R}$, define the z-slice $S_z := \{(x, y) \in \mathbb{R}^2 : (x, y, z) \in S\}$.

i. For which values of z is S_z non-empty? _____

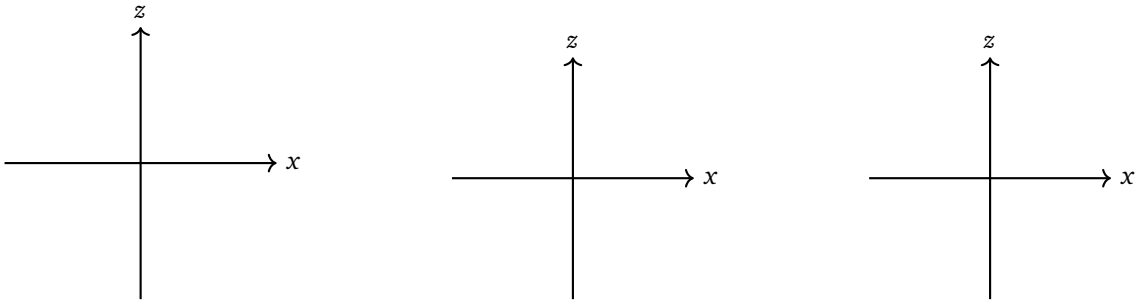
ii. Sketch the progression of z-slices from smaller to larger z-values.



(c) For $y \in \mathbb{R}$, define the y-slice $S_y := \{(x, z) \in \mathbb{R}^2 : (x, y, z) \in S\}$.

i. For which values of y is S_y non-empty? _____

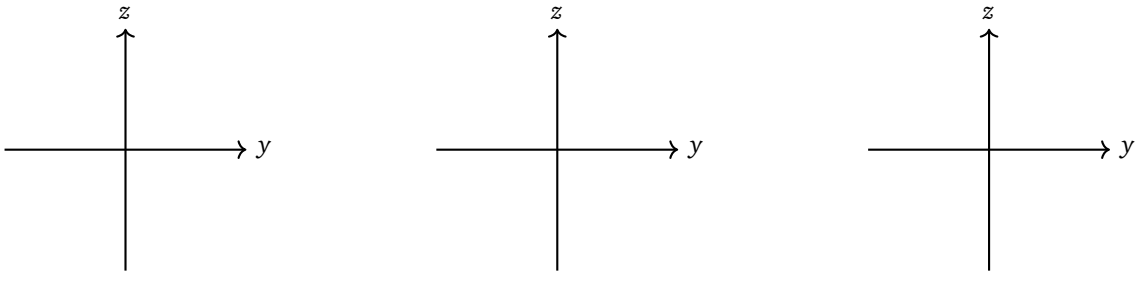
ii. Sketch a progression of y-slices from smaller to larger y-values.



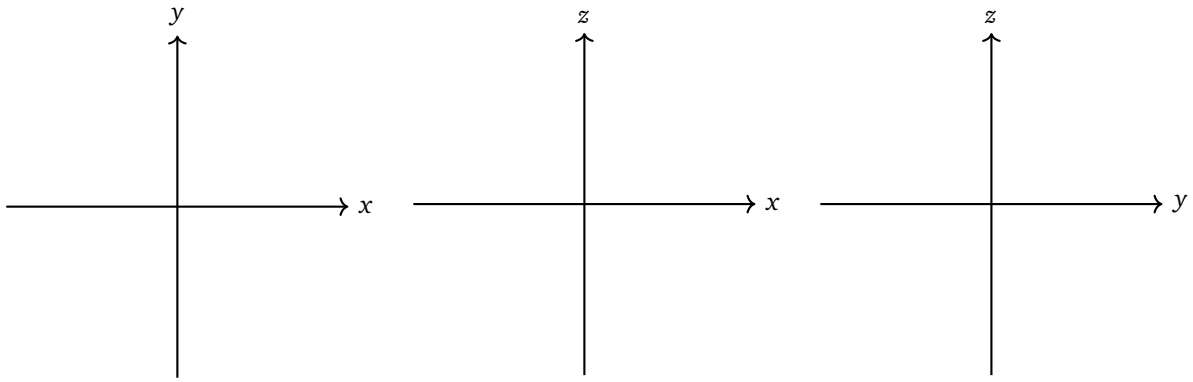
(d) For $x \in \mathbb{R}$, define the x-slice $S_x := \{(y, z) \in \mathbb{R}^2 : (x, y, z) \in S\}$.

i. For which values of x is S_x non-empty? _____

ii. Sketch a progression of x-slices from smaller to larger x-values.



(e) Sketch the three 2D projections of S into the (x,y)-plane, (x,z)-plane, and (y,z)-plane.



projection into (x,y)-plane | projection into (x,z)-plane | projection into (y,z)-plane

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T15.2 The solid S has continuous mass density $\delta(x,y,z) \geq 0$. You will express its mass $m = \int_S \delta \, dV$ as an iterated integral $dz\,dx\,dy$.

(a) To express its mass $dz\,dx\,dy$ using the *slicing* method, which slices should you use?

(b) To express its mass $dz\,dx\,dy$ using the *projection* method, which projection should you use?

(c) Use the projection method to express the mass of S as a triple iterated integral $dz\,dx\,dy$.

(d) How can you test the correctness of your expression from part (c)? Why is this test not necessarily perfect? Briefly explain in words. *Hint 1:* Volume and mass are related. *Hint 2:* Combine forces with another group.

T15.3 Now, you will express the mass of S using 2 other orders of integration. Include any rough work here.

(a) Use the slicing method to express the mass of S as one or more iterated integral(s) $dx\,dy\,dz$.

(b) Use any method to express the mass of S as one or more iterated integral(s) $dy\,dx\,dz$.

Hint: You will need at least 3 triple iterated integrals.

(c) **(Optional)** After tutorial, check your 3 expressions are correct by direct calculation of volume.